Multi-Tag Ranging and Localization: Evaluation

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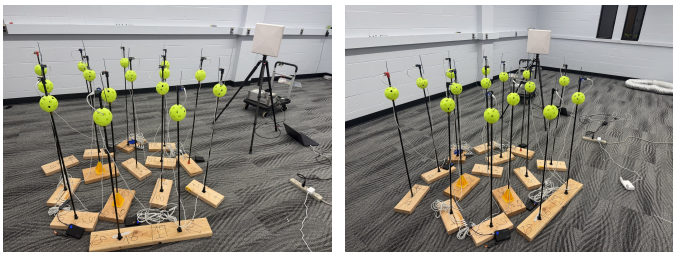


Fig. 1. Deployment of Exp. 10 (15 tags), viewed from two sides.

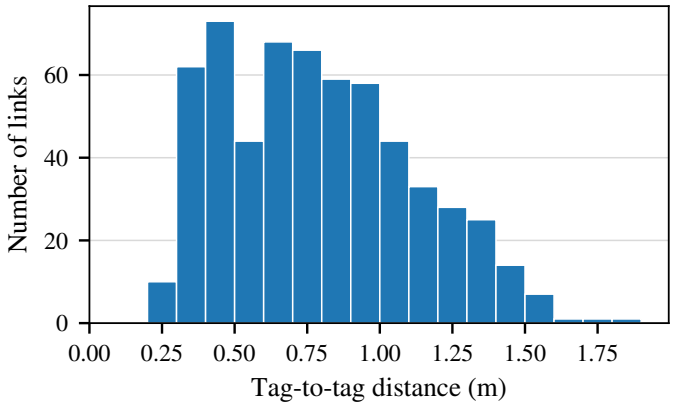


Fig. 2. Distribution of ground-truth tag-to-tag distances over all 909 links of Exp. 1–12 (bin width 0.1 m).

I. Evaluation Setup

A. Deployments

We evaluate on 12 indoor deployments (Exp. 1–12). Each deployment places 12 or 15 tags on vertical poles in an area with sides of 0.99 m to 1.75 m (Table I). Fig. 1 shows Exp. 10. Fig. 4 shows the layout of every deployment.

B. Ground truth

Tag positions are taken from a 3D scan of each deployment. Ground-truth link distances are the 3D distances between tags. Localization is evaluated in the horizontal (x, y) plane.

C. Measurements

Every tag pair (link) is measured at 18 carrier frequencies from 775 MHz to 955 MHz in 10 MHz steps (815 MHz excluded), giving a 180 MHz span. Each link is measured in 3 runs. All links are measured: 66 per deployment with 12 tags and 105 per deployment with 15 tags, 909 links and 2727 link measurements in total.

Link distances range from 0.21 m to 1.90 m, with a median of 0.73 m (Fig. 2).

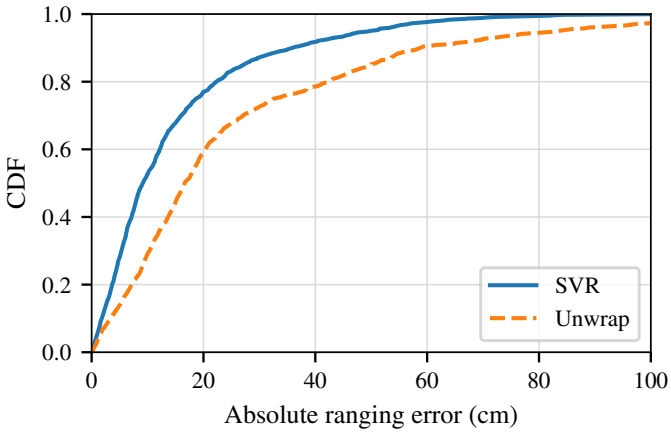


Fig. 3. CDF of absolute tag-to-tag ranging error, SVR vs. Unwrap, over 2727 link measurements (909 links × 3 runs) from Exp. 1–12. The x-axis is cut at 100 cm; 2.7% of Unwrap errors and 0.06% of SVR errors exceed it.

D. Ranging protocol

Ranging follows [1]. SVR is evaluated with leave-one-experiment-out cross-validation: the model for each experiment is trained on the other experiments only. A 13th deployment, not reported here, is used only as training data. Unwrap requires no training. Each of the 2727 link measurements is one test sample.

E. Localization protocol

Localization follows [2] and uses the SVR ranges as input. In each deployment, 6 tags are anchors with known positions; the remaining 6 or 9 tags are localized (81 tags in total). The anchors are the 6 tags whose convex hull has the largest area, so the anchors lie on the perimeter of the deployment.

II. Results

A. Ranging accuracy

Fig. 3 compares the absolute ranging error of SVR and Unwrap over all 2727 link measurements. SVR has a median error of 9.2 cm and a 90th-percentile error of 35.7 cm. Unwrap has a median error of 16.7 cm and a 90th-percentile error of 58.7 cm. SVR has the lower median error in every deployment (Table I).

B. Localization accuracy

Fig. 5 shows the localization error of the 81 localized tags. The median error is 11.0 cm, the mean is 13.4 cm, the 90th percentile is 22.0 cm, and the maximum is 54.4 cm. Per-deployment medians (Table I) range from 3.8 cm (Exp. 6) to 20.4 cm (Exp. 1). Fig. 4 shows the estimated position of every tag.

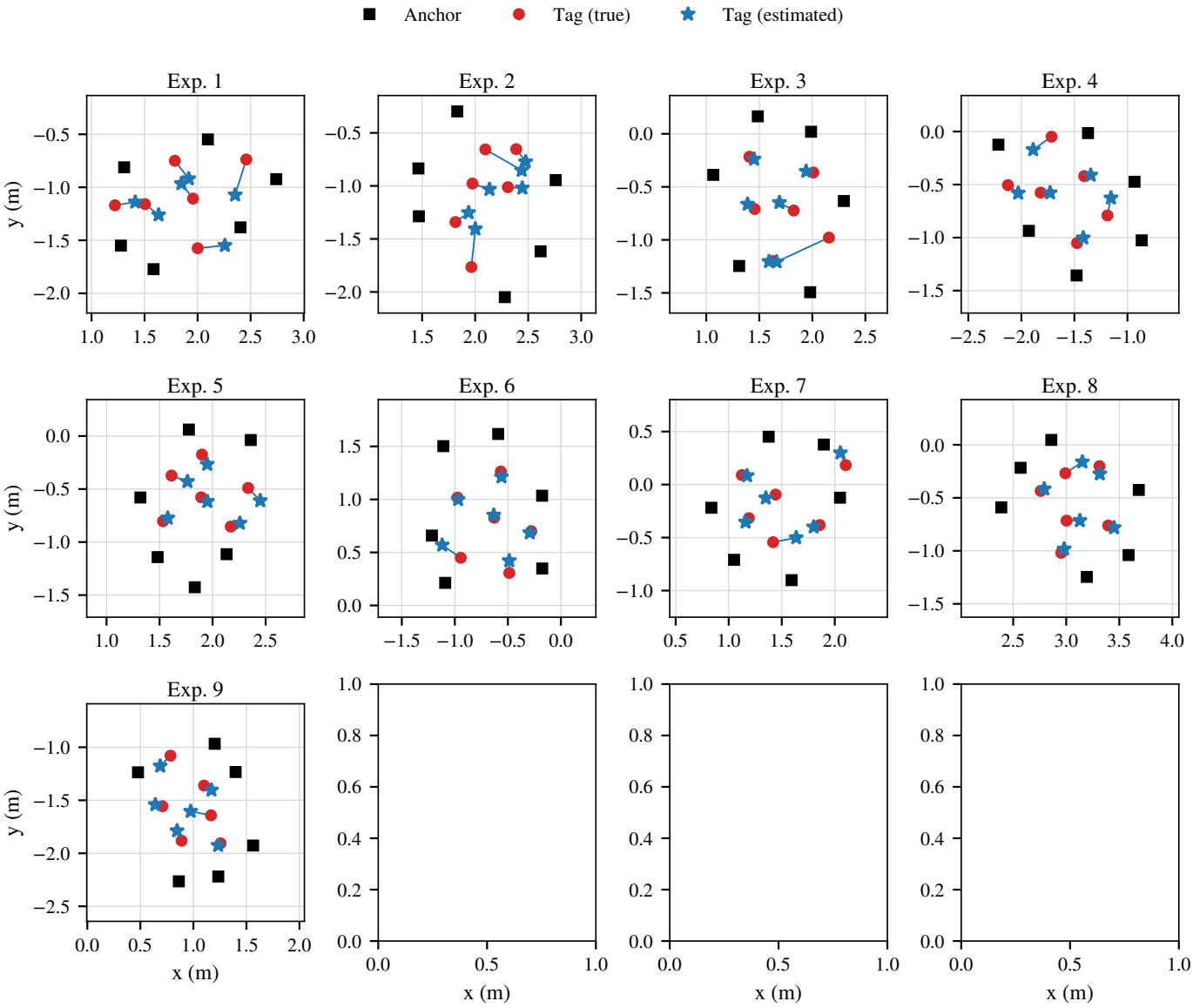


Fig. 4. Layouts of Exp. 1–12 in the horizontal plane. Squares: anchors. Circles: true positions of localized tags. Stars: positions estimated by the merged method; each line joins a tag's true and estimated positions. All panels use the same scale.

References

[1] M. Singh, Y. Xie, A. Ahmad, M. Stanaćević, S. R. Das, and P. M. Djurić, "Tag-to-tag range estimation for passive backscattering tags," in *Proc. 18th Int. Conf. Communication Systems and Networks (COMSNETS)*, Bengaluru, India, Jan. 2026, pp. 980–984, doi: 10.1109/COMSNETS67989.2026.11418103.

[2] M. Singh, Y. Xie, A. Ahmad, M. Stanaćević, S. R. Das, and P. M. Djurić, "Sub-decimeter range estimation and localization for passive backscattering tag-to-tag networks," unpublished manuscript, under revision for *ACM Trans. Internet of Things*, 2026.

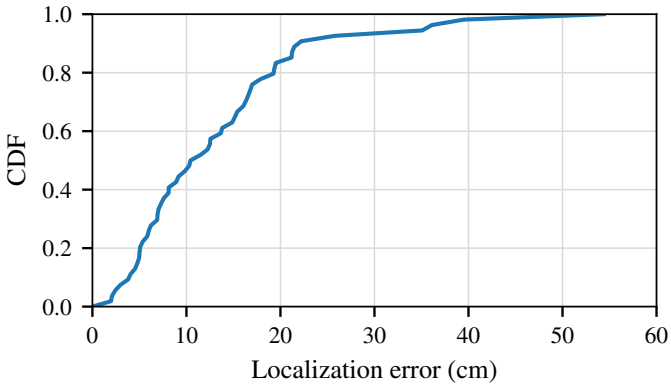


Fig. 5. CDF of 2D localization error of the merged method over the 81 localized tags of Exp. 1–12 (6 anchors per deployment).

TABLE I
DEPLOYMENTS AND PER-DEPLOYMENT MEDIAN ERRORS. DISTANCES ARE GROUND-TRUTH TAG-TO-TAG DISTANCES.

	Nodes				Link distance (m)			Extent (m)		Ranging error (cm)		Loc. error
Exp.	Tags	Anchors	Localized	Links	Min	Median	Max	x	y	SVR	Unwrap	(cm)
1	12	6	6	66	0.29	0.81	1.59	1.52	1.22	12.2	16.1	20.4
2	12	6	6	66	0.30	0.86	1.90	1.29	1.75	11.5	16.3	16.1
3	12	6	6	66	0.33	0.84	1.73	1.23	1.66	10.6	16.6	7.5
4	12	6	6	66	0.31	0.80	1.63	1.35	1.34	10.5	19.1	10.6
5	12	6	6	66	0.26	0.72	1.49	1.04	1.49	8.0	15.3	9.8
6	12	6	6	66	0.28	0.79	1.51	1.04	1.40	9.8	16.1	3.8
7	12	6	6	66	0.28	0.76	1.38	1.27	1.35	8.2	18.2	7.9
8	12	6	6	66	0.29	0.70	1.34	1.30	1.29	7.3	14.3	6.6
9	12	6	6	66	0.28	0.68	1.34	1.09	1.30	8.3	16.8	9.2
All	108	54	54	594	0.26	0.75	1.90			9.2	16.7	11.0